IN A LAND FAR, FAR AWAY...

"Finding a second Earth is not a matter of 'if', but of 'when'," says Thomas Zurbuchen of NASA. But, is it really so inevitable? Is life an all-pervading incident, or are we belittling its exclusivity with our latest discoveries?

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In recent times, not many intergalactic discoveries have received the amount of attention and craze as the TRAPPIST-1 system. The widespread fanfare of sorts was led by scientists who stated that the new solar system could be our best (and closest) bet to find life beyond planet Earth. This solar system had seven planets orbiting a tiny star in orbits in the 'habitable zone'. Quite understandably, everyone's excited, and also scared, mystified and keen to know if there really are more of us present in the expanding universe.

The question is, are we channelling our rationale rather blindly? In this issue, we look deeper into the discovery of the TRAPPIST-1 solar system, and try to reason crucial answers to niggling questions that have lingered at the back of our minds.

This is where you may get to live, some day

What is the TRAPPIST-1 discovery?

Using the Transiting Planets and Planetesimals Small Telescope (TRAPPIST) placed in La Silla Observatory, Chile, a team of scientists spotted three planets orbiting a dwarf star about 39 light years from Earth. This effort was led by astrophysicist Michael Gillion at the University of Liege, Belgium. The star itself was discovered by the Two Micron All-Sky Survey (2MASS) back in 1999, and finally, on February 22, 2017, four more planets were discovered in the system, bringing the total number of planets up to seven.

In terms of the TRAPPIST star's composition, it was observed and classified to be an "ultra cool" dwarf star with metallic constituency. Interestingly, all the planets that have been discovered in this solar system fall in a unique spatial distribution, and are much closer to each other than any of the planets in our solar system. This, as the discovery revealed, has significant consequences.

For one, this solar system is much younger than that of ours, aged about 500 million years. In contrast, our solar system is calculated to be 4.6 billion years old, and the very first signs of life appeared approximately 3.8 billion years ago. Interestingly, the TRAPPIST-1 solar system is younger than the 800 million years it took for us to support life.

The headlining discovery, however, is that three out of the seven discovered planets lie in the habitable zone. A solar system's habitable zone is a belt where rocky planets may exist with optimum heat, light and atmospheric conditions, all of which would lead to the planets being able to support forms of life as we know them. This is what made its discovery so celebrated, because we've never come across a solar system that has three Earth-like planets. It is also only 39 light years away from Earth, much closer than a number of other Earth-like exoplanets that have been discovered previously.

Gauging its significance

Seeing that the TRAPPIST star is significantly cooler than other stars (even our Sun), each of these exoplanets orbit very close to it, and each other.

This has led to NASA initiating further studies on how the forces interact among planets in this solar system. The planets are so close that if one looked up at the